

**Assessing the Incremental Predictive Value of Depressive Symptom Severity
in Healthy-Weight Prediction Using Machine Learning and Explainable
Artificial Intelligence**

Final Report

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Project Repository and Reproducibility

GitHub repository: <https://github.com/deepikashekhar/nhanes-healthy-weight-prediction>

The repository is intended to contain the NHANES source-file documentation, processed analytical data, data dictionary, executable notebook, exported tables, and figures used in this report. The analytical notebook used for this final report completed all reproducibility checks, including identical baseline/extended split membership, a fixed random seed, six evaluated model specifications, and confirmation that PHQ9_TOTAL was the only predictor added to the extended specification.

Abstract

Unhealthy weight is a multidimensional population-health problem influenced by demographic, socioeconomic, behavioural, dietary, sleep, and psychological factors. This study examined a specific unresolved question: whether depressive symptom severity contributes useful information for understanding and predicting healthy-weight status after conventional demographic and lifestyle information is already available. A reproducible statistical and machine-learning framework was developed using the 2017-2018 National Health and Nutrition Examination Survey (NHANES). The final analytical cohort comprised 4,657 U.S. adults aged 20 years or older; 1,134 (24.35%) were classified as healthy weight and 3,523 (75.65%) as not healthy weight. The binary outcome was derived from measured body mass index, while depressive symptom severity was represented by the validated nine-item Patient Health Questionnaire (PHQ-9). The baseline architecture contained 14 demographic, socioeconomic, activity, sedentary, sleep, and dietary predictors; the extended architecture added PHQ9_TOTAL as the sole additional predictor. Analyses were implemented in Python using pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost, SHAP, and Matplotlib on a CPU-based notebook environment. Logistic Regression, Random Forest, and XGBoost were evaluated using a fixed stratified 80/20 split, five-fold stratified cross-validation, and multiple held-out metrics. PHQ-9 was independently associated with healthy-weight status (adjusted OR = 0.9729 per point, 95% CI [0.9552, 0.9910], $p = .003$), but the effect was small. Adding PHQ-9 changed held-out ROC-AUC by only +0.0024 for Logistic Regression, +0.0024 for Random Forest, and -0.0002 for XGBoost; paired-bootstrap confidence intervals included zero for all three models. Baseline XGBoost achieved the highest held-out ROC-AUC (0.7109) and was selected, although calibration was poor (Brier score = 0.2127). SHAP showed that PHQ-9 could influence

individual predictions despite negligible incremental population-level discrimination. The framework is therefore best interpreted as a research and decision-support prototype rather than a clinically deployable risk calculator. It demonstrates that statistical association, feature contribution, and incremental predictive utility are distinct concepts and provides a reproducible basis for future longitudinal and externally validated research.

Assessing the Incremental Predictive Value of Depressive Symptom Severity in Healthy-Weight Prediction Using Machine Learning and Explainable Artificial Intelligence

Maintaining a healthy body weight is a major population-health challenge because overweight and obesity are associated with chronic disease, reduced quality of life, and substantial healthcare and socioeconomic burden (World Health Organization [WHO], 2024; Centers for Disease Control and Prevention [CDC], 2024). Weight status is not produced by a single behaviour or characteristic. Demographic and socioeconomic conditions, physical activity, sedentary behaviour, dietary intake, sleep, and psychological health interact in ways that require a multidimensional analytical approach (Hruby & Hu, 2015).

Psychological health is especially relevant because depression and obesity have been linked in longitudinal and meta-analytic research, including evidence of a bidirectional relationship (Luppino et al., 2010; Mannan et al., 2016; Pereira-Miranda et al., 2017). Yet association does not automatically imply predictive usefulness. A psychological measure may be statistically related to weight status while adding little new information once demographic and lifestyle variables are already represented. The present study was designed around that distinction.

The National Health and Nutrition Examination Survey (NHANES) is well suited to this question because it combines standardized examinations with demographic, socioeconomic, behavioural, dietary, sleep, and psychological information in a large U.S. population survey. Machine learning can capture nonlinear relationships in such multidimensional data, while explainable artificial intelligence (XAI) can clarify why a model produces particular predictions (Arrieta et al., 2020; Esteva et al., 2019; Lundberg & Lee, 2017). Stakeholders include healthcare

researchers, public-health agencies, healthcare organizations, clinicians, and policymakers interested in transparent, evidence-based prevention strategies.

Problem Statement

The objective of this study is to predict and explain Y = binary healthy-weight status for adult NHANES 2017-2018 participants using X = demographic, socioeconomic, physical-activity, sedentary, sleep, and dietary predictors, and to test whether depressive symptom severity measured by PHQ-9 adds incremental predictive information. Success is evaluated primarily by held-out ROC-AUC, supported by PR-AUC, balanced accuracy, sensitivity, specificity, precision, F1 score, statistical association tests, paired-bootstrap uncertainty, calibration measures, and SHAP explainability.

Purpose of the Study

The study develops an interpretable healthy-weight classification framework, quantifies the adjusted association between depressive symptom severity and healthy-weight status, tests the incremental predictive value of PHQ-9, compares Logistic Regression with Random Forest and XGBoost, and uses SHAP to explain the selected model and the role of PHQ-9. The study is associative and predictive rather than causal; it does not diagnose depression and does not claim that depressive symptoms cause unhealthy weight.

Research Questions

RQ1. To what extent is depressive symptom severity associated with healthy-weight status among adults in NHANES 2017-2018 after accounting for demographic and lifestyle factors?

RQ2. Does incorporating depressive symptom severity significantly improve prediction of healthy-weight status beyond demographic and lifestyle variables?

RQ3. Which statistical or machine-learning model provides the strongest predictive performance for healthy-weight status?

RQ4. Which factors contribute most strongly to healthy-weight predictions, and what role does depressive symptom severity play when model behaviour is explained with SHAP?

Contributions and Expected Value

The practical contribution is a transparent framework for examining healthy-weight status using routinely collected multidimensional health information. The technical contribution is a matched baseline-versus-PHQ-9 experimental design across three model families, leakage-safe preprocessing, held-out model comparison, paired-bootstrap incremental-value testing, threshold robustness analysis, calibration assessment, and global/local SHAP explanations. The stakeholder value lies less in immediate individual diagnosis than in showing how psychological information can be evaluated rigorously alongside conventional population-health predictors without confusing statistical significance with predictive improvement.

Literature Review

Literature Review Approach

The review was organized around four themes aligned with the research questions: multidimensional determinants of healthy weight, depression and weight status, predictive modelling and validation, and explainable AI. Peer-reviewed and authoritative sources were identified through Google Scholar, PubMed, ScienceDirect, IEEE Xplore, and official public-health sources using terms including obesity, healthy weight, BMI, depression, PHQ-9, NHANES, physical activity, diet, sleep, machine learning, XAI, SHAP, and healthcare

prediction. Priority was given to sources that justified predictor domains, validated PHQ-9, established depression-weight relationships, introduced the selected algorithms, or supported rigorous validation and explainability.

Why PHQ-9 Was Selected

PHQ-9 was selected for substantive, measurement, and experimental reasons. Substantively, longitudinal evidence indicates that depression and obesity are related in both directions, making depressive symptom severity a plausible psychological correlate of weight status (Luppino et al., 2010). Measurement-wise, the PHQ-9 is a validated nine-item instrument that provides a continuous symptom-severity score and established severity categories (Kroenke et al., 2001). Experimentally, NHANES 2017-2018 contains the nine depression-screening items required to derive PHQ9_TOTAL, and the final cohort contained complete PHQ-9 scores. This allowed PHQ-9 to be added as exactly one new variable to an otherwise identical baseline model, creating a direct test of incremental predictive value rather than an uncontrolled comparison of different feature sets.

Summary of Key Literature

Table 1

Summary of Key Literature and Research Linkages

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	RQ/Decision Link
Hruby & Hu (2015)	Obesity determinants	Population-level literature	Narrative epidemiological review	Obesity is multifactorial across biological, behavioural, socioeconomic, environmental, and demographic domains.	Supports multidimensional baseline predictors; RQ1-RQ3.
Luppino et al. (2010)	Depression and obesity	Longitudinal studies	Systematic review/meta-analysis	Evidence supports a bidirectional longitudinal depression-obesity relationship.	Primary substantive rationale for PHQ-9; RQ1-RQ2.

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	RQ/Decision Link
Mannan et al. (2016)	Depression and obesity	Prospective studies	Systematic review/meta-analysis	Associations vary across populations and sex.	Supports cautious, non-causal interpretation of RQ1.
Kroenke et al. (2001)	Depression measurement	Primary-care validation	Psychometric/diagnostic validation	PHQ-9 demonstrated reliability and validity as a screening and severity measure.	Justifies PHQ9_TOTAL and severity categories; RQ1-RQ2.
Esteva et al. (2019)	AI in healthcare	Healthcare AI literature	Narrative review	ML/deep learning can model complex health data but require rigorous validation.	Supports ML comparison and broader AI context; RQ3.
Yu et al. (2018)	AI in healthcare	Biomedical/clinical AI	Review/perspective	Healthcare AI offers value but requires evaluation and generalizability.	Supports predictive-healthcare framing; RQ2-RQ4.
Topol (2019)	Human-AI integration	Healthcare decision making	Review/perspective	AI should augment human decision making and requires trust and interpretation.	Supports decision-support framing; RQ4.
Breiman (2001)	Ensemble learning	Benchmark datasets	Random Forest	Randomized tree ensembles can improve robustness and predictive accuracy.	Justifies Random Forest; RQ3.
Chen & Guestrin (2016)	Gradient boosting	Benchmark/applied datasets	XGBoost	Regularized scalable boosting achieves strong predictive performance.	Justifies XGBoost; RQ3.
Kohavi (1995)	Model validation	ML datasets	Cross-validation study	Validation design materially affects reliable model selection.	Supports five-fold CV plus held-out test; RQ2-RQ3.
Powers (2011)	Classification metrics	General classification	Metric review	Accuracy alone can mislead under class imbalance.	Supports ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, precision and F1.
Arrieta et al. (2020)	Explainable AI	High-stakes AI	Review/taxonomy	XAI supports transparency and accountability in consequential applications.	Conceptual foundation for RQ4.
Lundberg & Lee (2017)	Model interpretability	General ML	SHAP	SHAP assigns theoretically grounded local and global feature contributions.	Methodological basis for RQ4.

Note. The matrix links each source directly to a project decision or research question. Complete references are provided in the References section.

Thematic Synthesis

The literature establishes healthy weight as a multifactorial outcome. Epidemiological evidence supports including age, sex, race/ethnicity, socioeconomic position, physical activity, sedentary behaviour, diet, and sleep rather than reducing the problem to calorie intake alone (Chaput et al., 2020; Hruby & Hu, 2015; Mozaffarian et al., 2011; Swift et al., 2014). These domains therefore formed the baseline against which PHQ-9 was tested.

Depression-related literature supports the plausibility of a psychological contribution while also warning against causal overstatement. Meta-analyses report relationships between depression and overweight/obesity, but heterogeneity and bidirectionality mean that a cross-sectional NHANES analysis can establish association, not temporal direction (Luppino et al., 2010; Mannan et al., 2016; Pereira-Miranda et al., 2017). This motivated separate treatment of RQ1 association and RQ2 predictive increment.

Predictive-method literature supports comparing a transparent statistical benchmark with nonlinear ensemble methods. Logistic Regression provides interpretable adjusted odds ratios; Random Forest captures nonlinearities through ensembles of randomized trees; XGBoost uses regularized gradient boosting and is often effective for structured tabular data (Breiman, 2001; Chen & Guestrin, 2016; Hosmer et al., 2013). Deep-learning literature was reviewed as part of the broader healthcare-AI landscape, but deep neural networks were not used because the project involved a moderate-sized structured tabular dataset and the approved scope prioritized interpretable statistical and tree-based models (Esteva et al., 2019; Shickel et al., 2018).

The study also distinguishes prediction from interpretability. Strong discrimination does not by itself explain a prediction, and feature importance does not establish causality. SHAP was therefore used after model selection to identify global and individual contributions (Arrieta et al., 2020; Lundberg & Lee, 2017). Agentic AI or a large language model was not implemented in the final analytical pipeline; introducing one would add an unvalidated layer not required to answer the four approved research questions. A future dissertation could investigate a carefully governed language-model layer for translating validated model explanations into user-facing narratives, but that extension is outside the evidentiary claims of the present capstone.

Research Gap

The literature supports both a depression-weight relationship and the usefulness of machine learning in health analytics, but it does not resolve whether a validated depressive-symptom score adds meaningful predictive information after a broad demographic and lifestyle baseline is already represented. The present study addresses that gap with matched baseline and PHQ-9-extended models, formal uncertainty around changes in held-out ROC-AUC, and explainability analysis that separates population-level predictive increment from individual feature contribution.

Materials and Methods

Study Design and Data Source

This secondary, cross-sectional study used public-use NHANES 2017-2018 data from the National Center for Health Statistics (NCHS), CDC. Seven component files were merged using the unique participant identifier SEQN: Demographics (DEMO_J), Body Measures (BMX_J), Depression Screener (DPQ_J), Physical Activity (PAQ_J), Sleep Disorders (SLQ_J), Dietary Interview-Day 1 Total Nutrient Intakes (DR1TOT_J), and Weight History (WHQ_J). The unit of

analysis was an individual adult NHANES participant. The source data are de-identified public-use files. NHANES survey weights were not incorporated because the primary objective was model development and comparative prediction rather than nationally representative prevalence estimation; this decision limits population-level generalization.

Cohort Construction and Outcome

The merged raw dataset contained 9,254 participants. Restriction to age 20 years or older yielded 5,569 participants; excluding pregnancy reduced the cohort to 5,514; 5,120 had valid measured BMI; 4,729 had complete PHQ-9; and the final analytical cohort after excluding underweight participants comprised 4,657 adults. HEALTHY_WEIGHT was defined as 1 for BMI 18.5 to <25 kg/m² and 0 for BMI ≥25 kg/m². Participants with BMI <18.5 kg/m² were excluded because the research question contrasted healthy weight with overweight/obesity. BMI and BMI-derived variables were excluded from predictors to prevent target leakage.

Table 2

Final Analytical Cohort Characteristics

Characteristic	Final Value
Time period	NHANES 2017-2018
Unit of analysis	Individual adult participant
Final analytical N	4,657
Healthy weight	1,134 (24.35%)
Not healthy weight	3,523 (75.65%)
Target	HEALTHY_WEIGHT
Baseline predictors	14
Extended predictors	15; baseline + PHQ9_TOTAL

Note. HEALTHY_WEIGHT = 1 for measured BMI 18.5 to <25 kg/m² and 0 for BMI ≥25 kg/m²; participants with BMI <18.5 kg/m² were excluded.

Data Dictionary and Predictor Architecture

Table 3

Baseline and Extended Predictor Architecture

Variable	Definition	Type	Role
RIDAGEYR	Age at screening	Continuous	Baseline
RIAGENDR	Sex	Categorical	Baseline
RIDRETH3	Race/Hispanic origin	Categorical	Baseline
DMDEDUC2_CLEAN	Educational attainment	Categorical	Baseline
DMDMARTL_CLEAN	Marital status	Categorical	Baseline
INDFMPIR	Family income-to-poverty ratio	Continuous	Baseline
PAQ605_CLEAN	Vigorous work activity	Categorical	Baseline
PAQ620_CLEAN	Moderate work activity	Categorical	Baseline
PAQ650_CLEAN	Vigorous recreational activity	Categorical	Baseline
PAQ665_CLEAN	Moderate recreational activity	Categorical	Baseline
PAD680_CLEAN	Sedentary minutes/day	Continuous	Baseline
SLD012	Typical sleep duration	Continuous	Baseline
DR1TKCAL	Day-1 total energy intake	Continuous	Baseline
DR1TFIBE	Day-1 dietary fibre	Continuous	Baseline
PHQ9_TOTAL	Depressive symptom severity, 0-27	Discrete numeric	Extended only

Note. A fuller analytical data dictionary, including source and target variables and missing-data treatment, is provided in Appendix A.

Data Cleaning and Missing Data

Integrity checks covered participant identifiers, duplicates, target validity, PHQ-9 range, missingness, special NHANES response codes, and target leakage. Non-substantive codes were converted to missing for education, marital status, activity variables, and sedentary time.

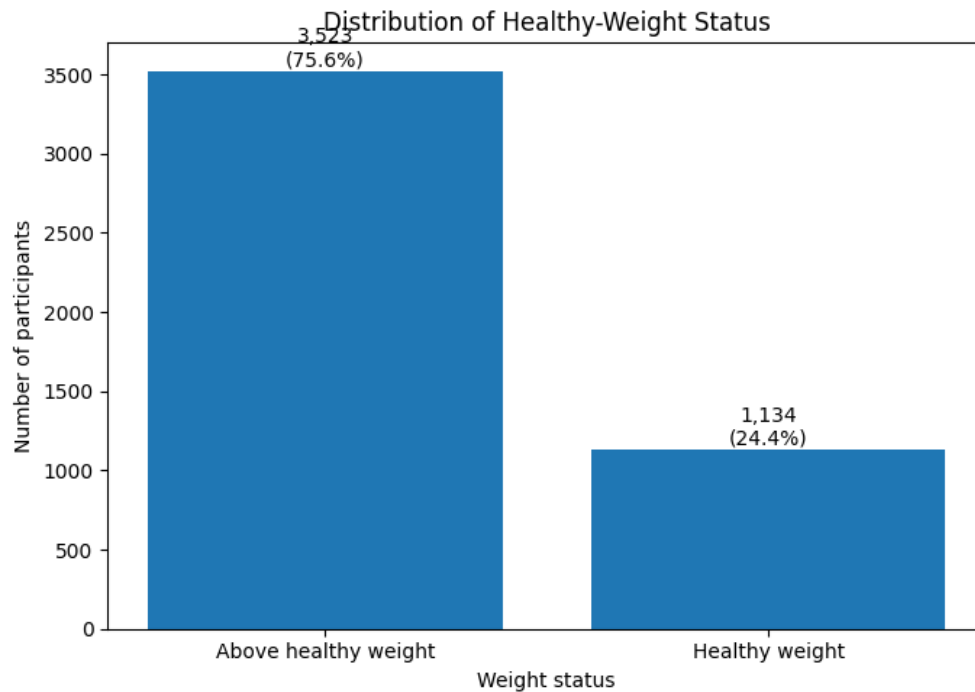
Complete-case analysis would have removed 811 participants (17.41%), so predictive models retained the full analytical cohort and handled predictor missingness inside training-only preprocessing pipelines. Continuous variables used median imputation; categorical variables

used most-frequent imputation. PHQ9_TOTAL and HEALTHY_WEIGHT had no missing values in the final cohort.

Exploratory multicollinearity analysis identified strong redundancy among dietary energy and macronutrient variables, including $VIF = 31.41$ for DR1TKCAL in the broader candidate set. A parsimonious energy-plus-fibre specification was retained; after refinement, the maximum VIF was 1.43. Protein, carbohydrate, and total-fat intake were removed from the final predictor set. This reduced redundancy while preserving total energy and dietary fibre as complementary dietary indicators.

Exploratory Data Analysis

The final cohort had mean age 51.41 years ($SD = 17.38$) and mean PHQ-9 score 3.21 ($SD = 4.24$; median = 2). PHQ-9 was strongly right-skewed. Healthy weight represented only 24.35% of the sample, so accuracy alone was considered insufficient. Exploratory screening showed that race/ethnicity had the largest categorical association with the target (Cramer's $V = 0.205$), while age had the largest absolute point-biserial association among continuous variables ($r = -0.080$). PHQ-9 had a small negative point-biserial association ($r = -0.057$).

Figure 1*Healthy-Weight Outcome Distribution*

Note. The figure reproduces the executed notebook output and shows the class imbalance:

24.35% healthy weight versus 75.65% not healthy weight. This motivated stratification and multiple evaluation metrics.

Research Hypotheses and Statistical Tests**Table 4***Research Questions, Hypotheses, Statistical Methods, and Decisions*

RQ	Null Hypothesis	Alternative Hypothesis	Method	Decision at alpha=.05 / Interpretation
RQ1	H0: After adjustment, PHQ-9 is not associated with healthy-weight status ($\beta_{\text{PHQ9}} = 0$; OR = 1).	Ha: PHQ-9 is associated with healthy-weight status ($\beta_{\text{PHQ9}} \neq 0$; OR $\neq 1$).	Welch t-test, Mann-Whitney U, chi-square, adjusted logistic regression	Reject H0 for association; adjusted p = .0034, but effect is small.

RQ	Null Hypothesis	Alternative Hypothesis	Method	Decision at $\alpha=.05$ / Interpretation
RQ2	H0: Adding PHQ-9 does not improve out-of-sample discrimination beyond the baseline predictors (Delta ROC-AUC = 0).	Ha: Adding PHQ-9 improves out-of-sample discrimination (Delta ROC-AUC > 0).	Matched baseline/extended models; paired bootstrap 95% CI for Delta ROC-AUC	Fail to reject H0 across all three model families; every CI includes zero.
RQ3	H0: No evaluated model demonstrates superior held-out discrimination under the prespecified ROC-AUC selection criterion.	Ha: At least one evaluated model demonstrates superior held-out discrimination.	Six-model held-out comparison after five-fold CV/tuning	Baseline XGBoost has the highest observed held-out ROC-AUC (0.7109) and is selected; no formal pairwise significance claim is made.
RQ4	H0: Model explanations do not reveal meaningful variation in feature contributions, including PHQ-9 in the extended model.	Ha: SHAP reveals heterogeneous feature contributions and a measurable explanatory role for PHQ-9.	Global and local SHAP	Descriptive/explainability hypothesis supported: contributions vary; PHQ9_TOTAL ranks 7th in extended XGBoost. SHAP is not a causal test.

Note. H0 = null hypothesis; Ha = alternative hypothesis. The prespecified significance level was $\alpha = .05$. Predictive research questions were evaluated primarily with held-out performance and uncertainty rather than p values alone.

Minimum Sample-Size Adequacy

Power and precision calculations were completed before final modelling. RQ1 used a two-proportion power analysis ($\alpha = .05$, power = .80, Cohen's $h = .20$) requiring 785 participants. RQ2 used an incremental-effect calculation ($\alpha = .05$, power = .80, Cohen's $f^2 = .02$, one tested predictor/block) requiring 395. RQ3 and RQ4 used 95% confidence-interval precision calculations requiring 385 each. The study-level minimum was therefore 785 participants; the available $N = 4,657$ exceeded the largest requirement by 3,872 participants.

Table 5

Minimum Sample-Size Requirement by Research Question

Research Question	Method	Key Parameters	Minimum N
RQ1	Two-proportion power	$\alpha=.05$; power=.80; $h=.20$; two-sided	785

Research Question	Method	Key Parameters	Minimum N
RQ2	Incremental model contribution	alpha=.05; power=.80; f^2 =.02; one tested block	395
RQ3	Confidence-interval precision	95% CI; e=.05; p=.50	385
RQ4	Confidence-interval/validated sample	95% CI; e=.05; p=.50	385

Note. The largest prespecified minimum requirement was below the final analytical sample of $N = 4,657$, supporting sample-size adequacy for the planned analyses.

Train-Test Split and Leakage-Safe Preprocessing

A fixed stratified 80/20 split with `random_state = 42` produced 3,725 training participants and 932 held-out test participants while preserving the target distribution. Baseline and extended models used identical participant rows; `PHQ9_TOTAL` was the only added predictor. Preprocessors were fitted on training data only. Six continuous baseline variables were median-imputed and standardized; eight categorical variables were mode-imputed and one-hot encoded with a reference category dropped and unknown test categories ignored. The extended pipeline added `PHQ9_TOTAL` as a seventh continuous variable. The transformed matrices contained 25 baseline features and 26 extended features, with no missing values after preprocessing.

Model Development and Evaluation

Three model families were prespecified: Logistic Regression as the statistical and interpretable benchmark, Random Forest as a bagged nonlinear ensemble, and XGBoost as a regularized boosted-tree model. Five-fold stratified cross-validation on the training set supported tuning and comparison without exposing the held-out test set. Final selection used held-out ROC-AUC as the primary criterion. Because the positive healthy-weight class was the minority, PR-AUC, balanced accuracy, sensitivity, specificity, precision, and F1 were reported as complementary metrics. The final XGBoost model was additionally examined for threshold

sensitivity and probability calibration using the Brier score, Brier Skill Score, and calibration curves.

Software and Compute Environment

The final reproducibility cell recorded Python 3.13.15, NumPy 2.1.3, pandas 2.2.3, scikit-learn 1.6.1, XGBoost 3.4.1, SHAP 0.52.0, SciPy 1.16.3, and Matplotlib 3.10.0 on a Linux x86_64 CPU environment. GPU acceleration was not required for the structured tabular models used in this capstone.

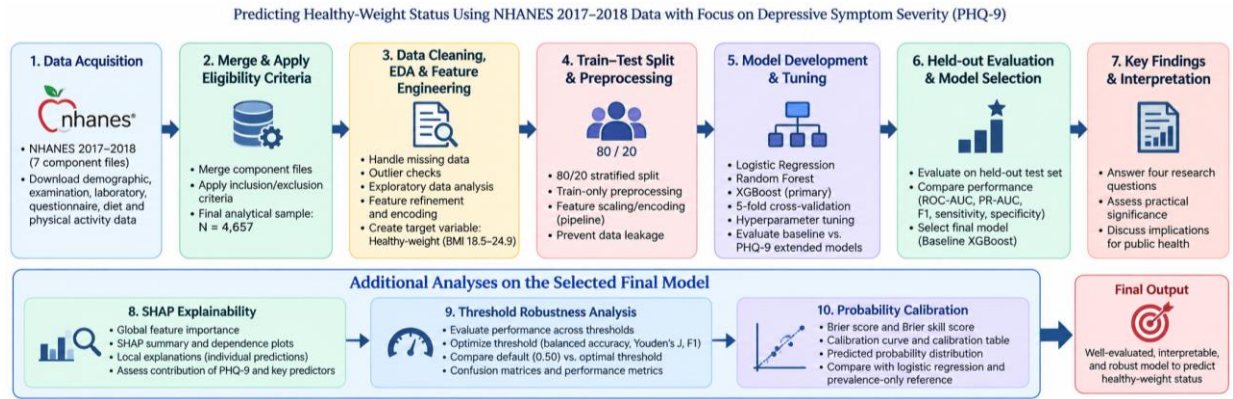
Architecture and Workflow

System Overview

The workflow was designed to preserve traceability from public NHANES source files to final research conclusions. Raw component files were merged and filtered to construct the analytical cohort; quality checks and exploratory analysis informed feature refinement; baseline and PHQ-9-extended pipelines were split before preprocessing; models were tuned on training data; the held-out test set was used for final comparison; SHAP, threshold robustness, and calibration analyses were then applied to interpret and qualify the selected result.

Figure 2

End-to-End Analytical Workflow



Note. The diagram summarizes the complete executed workflow from NHANES ingestion through model evaluation and post-selection explainability/robustness. Preprocessing parameters are learned only from the training data.

Workflow Components

Data ingestion used NHANES XPT files merged by SEQN. Preprocessing converted non-substantive response codes to missing, used training-derived imputation, standardized continuous predictors, and one-hot encoded categorical predictors. EDA evaluated distributions, missingness, class balance, associations, correlations, and multicollinearity. Feature engineering created PHQ9_TOTAL and HEALTHY_WEIGHT and selected the final 14-predictor baseline. Model development compared Logistic Regression, Random Forest, and XGBoost under matched baseline and extended specifications. Model evaluation used held-out discrimination and classification metrics, followed by paired-bootstrap incremental-value analysis, SHAP, threshold robustness, and calibration. Deployment was not performed; the proposed implementation is discussed as a future validated decision-support workflow rather than as a current clinical product.

Results

RQ1: Association Between PHQ-9 and Healthy-Weight Status

Participants classified as not healthy weight had a higher mean PHQ-9 score ($M = 3.347$, $SD = 4.354$) than healthy-weight participants ($M = 2.788$, $SD = 3.824$). Welch's t-test was significant, $t = -4.1312$, $p = .000037$, and the Mann-Whitney U test was also significant, $U = 1,866,598$, $p = .000662$. The effect was nevertheless small: Cohen's $d = -0.1320$ and rank-biserial correlation = -0.0656 .

PHQ-9 severity categories showed an overall downward pattern in healthy-weight prevalence: 25.87% for minimal symptoms, 20.39% mild, 19.85% moderate, 20.00% moderately severe, and 10.00% severe. The association was significant, $\chi^2(4) = 19.4029$, $p = .000655$, but Cramer's $V = 0.0645$ indicated a very small relationship. In the adjusted Logistic Regression controlling for the prespecified demographic, socioeconomic, activity, sedentary, sleep, and dietary predictors, PHQ9_TOTAL remained significant: adjusted OR = 0.9729, 95% CI [0.9552, 0.9910], $p = .003407$. Each additional PHQ-9 point was therefore associated with approximately 2.71% lower adjusted odds of healthy-weight status, holding other predictors constant.

RQ2: Incremental Predictive Value of PHQ-9

Table 6

Incremental Predictive Value of PHQ-9 Across Model Families

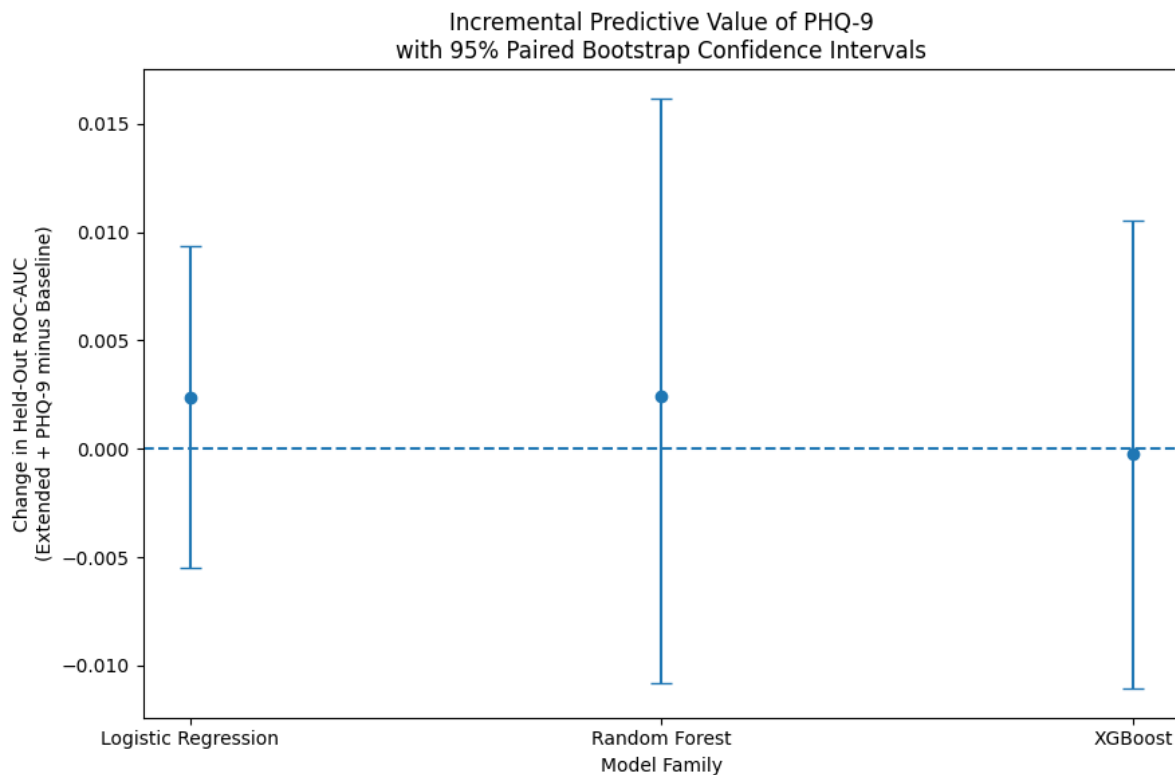
Model	Base ROC	Ext. ROC	Delta ROC	95% Bootstrap CI	Base PR	Ext. PR	Delta PR
Logistic Regression	0.6838	0.6862	+0.0024	[-0.0055, 0.0094]	0.4331	0.4355	+0.0023
Random Forest	0.7068	0.7092	+0.0024	[-0.0108, 0.0162]	0.4553	0.4605	+0.0052
XGBoost	0.7109	0.7107	-0.0002	[-0.0110, 0.0105]	0.4477	0.4494	+0.0017

Note. Delta values equal extended minus baseline performance. All paired-bootstrap 95% confidence intervals for change in ROC-AUC included zero.

Across all model families, the absolute change in ROC-AUC was no larger than 0.0024. Every paired-bootstrap 95% confidence interval for the change in ROC-AUC included zero. The evidence therefore does not support meaningful incremental improvement in out-of-sample discrimination from adding PHQ-9, even though PHQ-9 was independently associated with the outcome under RQ1.

Figure 3

Incremental Predictive Value of PHQ-9



Note. All three paired-bootstrap confidence intervals cross zero. This is the clearest visual evidence that the small observed ROC-AUC changes are compatible with sampling variation.

RQ3: Comparative Model Performance and Selection

Table 7

Held-Out Performance Across Six Model Specifications

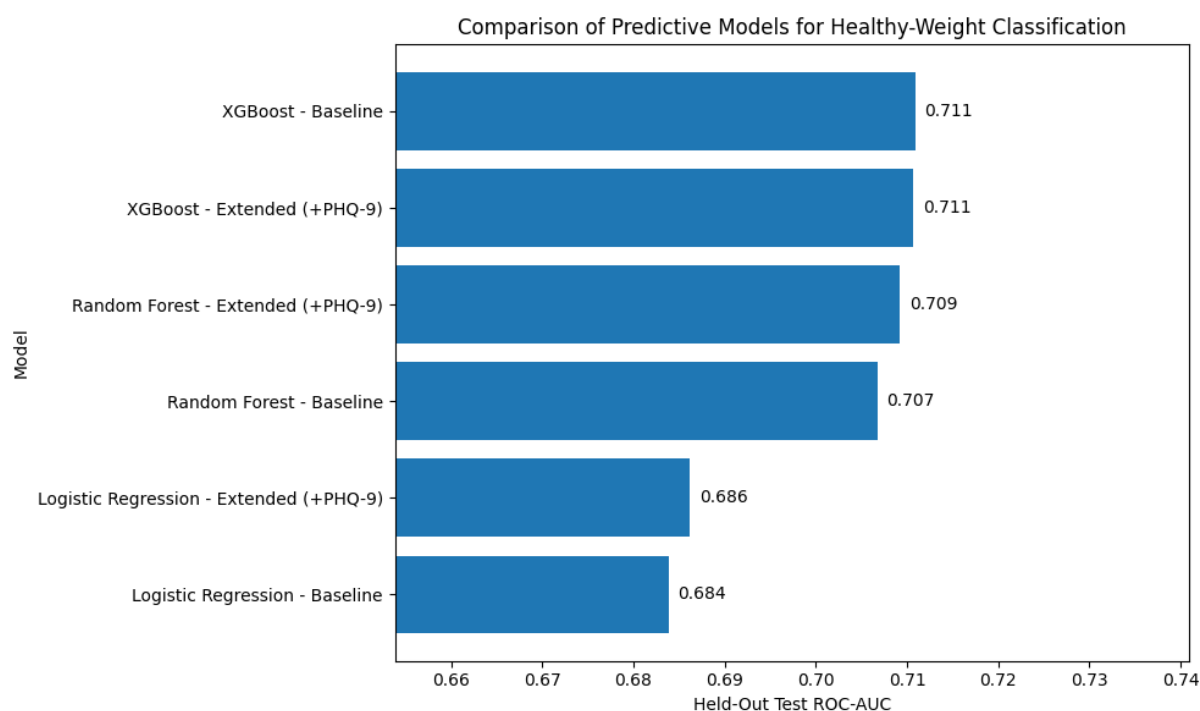
Model	Specification	ROC-AUC	PR-AUC	Accuracy	Bal. Acc.	Sensitivity	Specificity	Precision	F1
XGBoost	Baseline	0.7109	0.4477	0.7028	0.6572	0.5683	0.7461	0.4188	0.4822
XGBoost	Extended + PHQ-9	0.7107	0.4494	0.7017	0.6580	0.5727	0.7433	0.4180	0.4833
Random Forest	Extended + PHQ-9	0.7092	0.4605	0.7704	0.5809	0.2115	0.9504	0.5783	0.3097
Random Forest	Baseline	0.7068	0.4553	0.7629	0.5774	0.2159	0.9390	0.5326	0.3072
Logistic Regression	Extended + PHQ-9	0.6862	0.4355	0.7629	0.5341	0.0881	0.9801	0.5882	0.1533
Logistic Regression	Baseline	0.6838	0.4331	0.7618	0.5319	0.0837	0.9801	0.5758	0.1462

Note. Metrics are from the untouched held-out test set ($N = 932$) at threshold 0.50. Baseline

XGBoost was selected using the prespecified primary criterion of highest held-out ROC-AUC.

Figure 4

Held-Out ROC-AUC Comparison Across Six Model Specifications

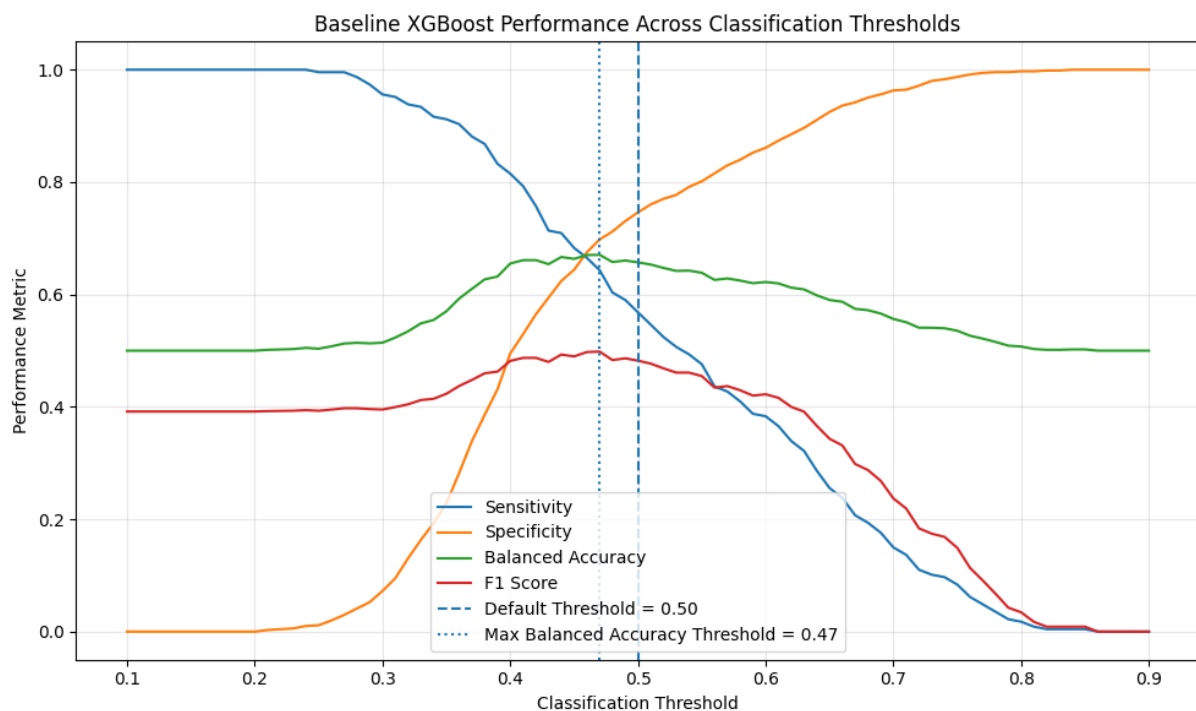


Note. Baseline XGBoost achieved the highest observed held-out ROC-AUC (0.7109), narrowly exceeding extended XGBoost and the Random Forest specifications.

At threshold 0.50, selected baseline XGBoost produced 526 true negatives, 179 false positives, 98 false negatives, and 129 true positives. A secondary post-hoc threshold of 0.47 increased balanced accuracy from 0.6572 to 0.6705 and sensitivity from 0.5683 to 0.6432, while specificity declined from 0.7461 to 0.6979. The independently derived Youden threshold (0.4651) was similar. These results demonstrate an operating-threshold trade-off; they do not establish a prospectively validated threshold.

Figure 5

Threshold Robustness for Baseline XGBoost

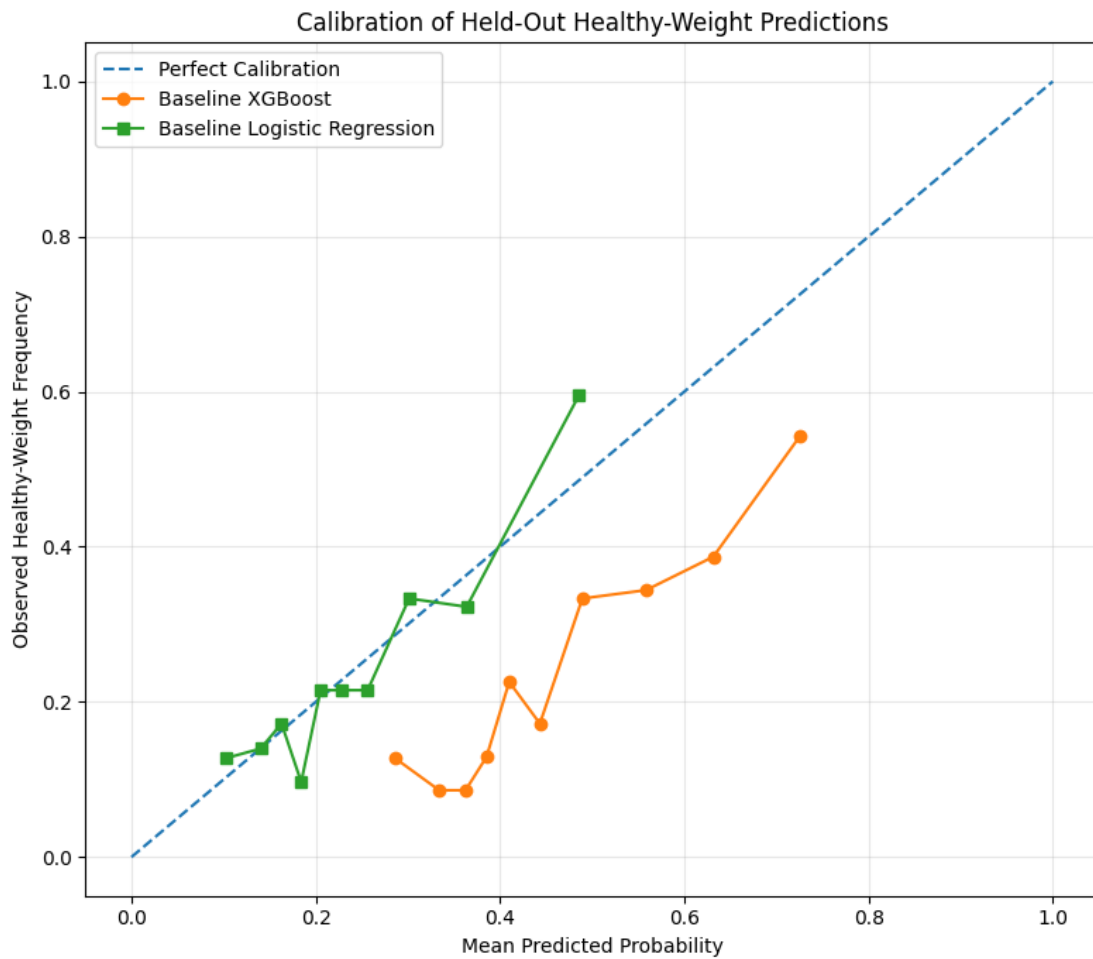


Note. The executed notebook evaluated performance across thresholds. A threshold near 0.47 improved balance and sensitivity but reduced specificity, illustrating why deployment thresholds must be selected for a defined use case.

Calibration qualified the model-selection result. Baseline XGBoost had Brier score = 0.2127, Brier Skill Score = -0.1547, and mean absolute calibration gap = 0.2192. A prevalence-only reference had Brier score = 0.1842, while baseline Logistic Regression had Brier score = 0.1678 and Brier Skill Score = 0.0890. Thus, XGBoost discriminated better but its raw probabilities were poorly calibrated and should not be interpreted directly as individual risk probabilities.

Figure 6

Calibration of Held-Out Healthy-Weight Predictions



Note. The XGBoost calibration curve lies systematically below the perfect-calibration line, consistent with overprediction of healthy-weight probability in many bins. Logistic Regression is visibly better calibrated.

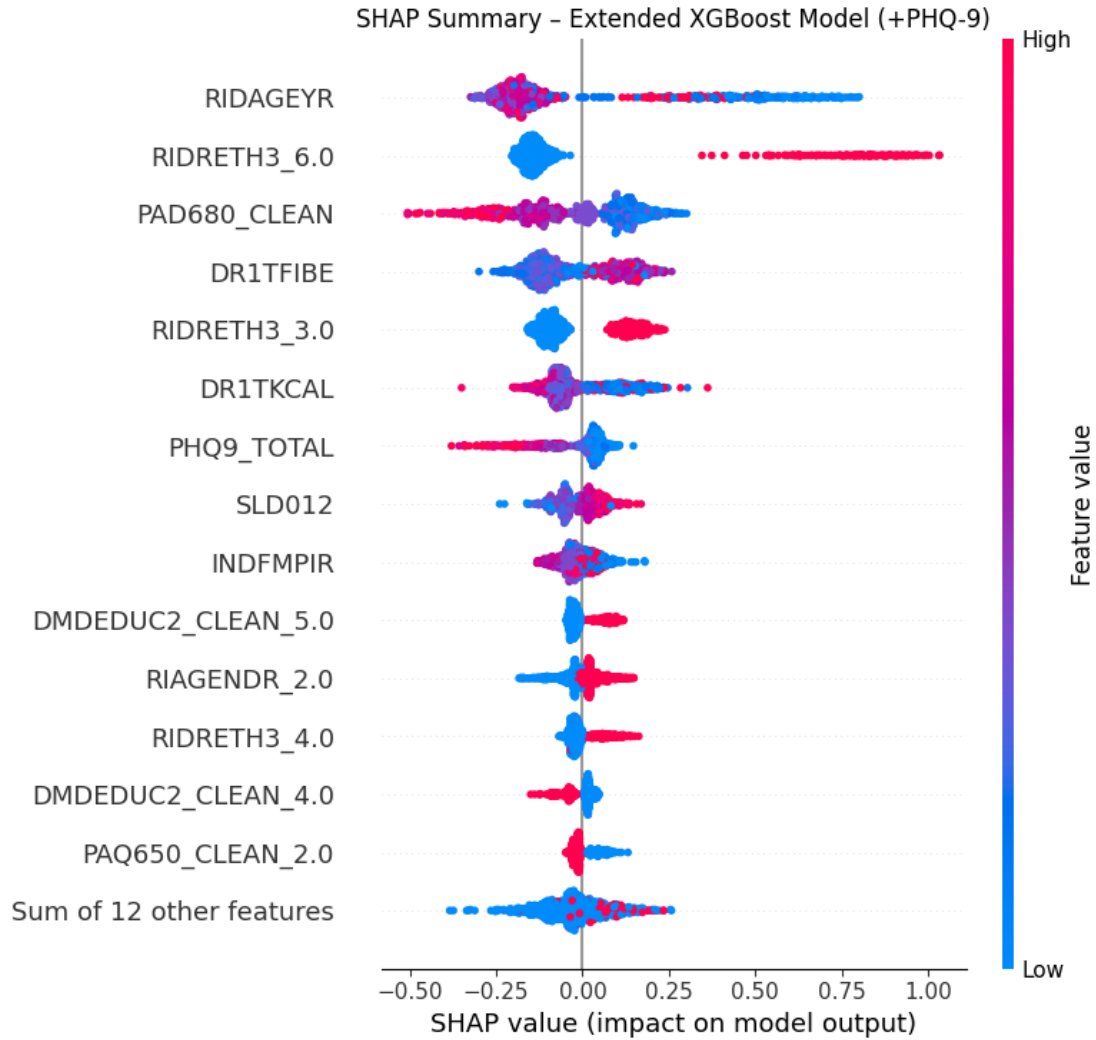
RQ4: Explainability and the Role of PHQ-9

Global SHAP analysis of baseline XGBoost ranked age (RIDAGEYR) first by mean absolute SHAP (0.2554), followed by race/ethnicity category RIDRETH3_6.0 (0.2384), sedentary time (PAD680_CLEAN; 0.1475), RIDRETH3_3.0 (0.1119), dietary fibre (0.1079), total energy intake (0.0835), and sleep duration (0.0466). This pattern confirms that the selected model uses multiple demographic and lifestyle signals rather than one dominant variable.

Because PHQ-9 is absent from the selected baseline model, a supplementary SHAP analysis was conducted on extended XGBoost specifically to examine depressive symptom severity. PHQ9_TOTAL ranked seventh with mean absolute SHAP = 0.0687. Higher PHQ-9 values generally shifted model output away from healthy-weight status, directionally consistent with the adjusted Logistic Regression association. The important nuance is that a variable can influence individual predictions without improving aggregate discrimination when added to an already informative feature set.

Figure 7

SHAP Summary for Extended XGBoost Including PHQ-9



Note. PHQ9_TOTAL appears seventh in global importance. Higher PHQ-9 values generally contribute negative SHAP values, shifting predictions away from healthy-weight status. SHAP values describe model contribution, not causality.

Local SHAP explanations were generated for two representative held-out participants. One participant had predicted healthy-weight probability 0.8549 and actual class 1; another had predicted probability 0.2007 and actual class 0. These examples demonstrate that the direction and magnitude of feature contributions differ across individuals. The notebook screenshots for both local explanations are reproduced in Appendix C.

Research Question Conclusions

Table 8

Final Answers to the Four Research Questions

RQ	Final Answer
RQ1	Yes. PHQ-9 is independently associated with healthy-weight status after adjustment, but the effect is small (OR = 0.9729 per point; $p = .0034$).
RQ2	No meaningful incremental predictive gain was established. Delta ROC-AUC was approximately zero across all three model families and all paired-bootstrap CIs included zero.
RQ3	Baseline XGBoost achieved the highest held-out ROC-AUC (0.7109) and was selected for discrimination, but its raw probabilities were poorly calibrated.
RQ4	SHAP shows that predictions reflect multiple demographic/lifestyle factors. In extended XGBoost, PHQ-9 ranks seventh and can affect individual predictions despite negligible incremental discrimination.

Note. Conclusions distinguish statistical association, incremental predictive value, comparative predictive performance, and explainability.

Discussion and Practical Significance

Overall Interpretation: What Did the Study Find?

The central finding is not simply that depression is related to weight. The stronger contribution is the empirical separation of three concepts that are often conflated: statistical association, predictive utility, and explainability. PHQ-9 showed a statistically significant independent association with healthy-weight status. It also had a visible role in the extended XGBoost SHAP explanation. Yet it did not materially improve held-out ROC-AUC beyond the demographic and lifestyle baseline. The study therefore establishes that a variable may be scientifically relevant without being a strong incremental predictive feature.

This finding answers the project's core research problem. The evidence supports the proposition that psychological well-being is one component of a multifactorial healthy-weight context, but it does not support the stronger proposition that PHQ-9 materially improves healthy-

weight classification when conventional demographic and lifestyle information is already available. In this sense, the research hypothesis receives partial rather than complete support.

Did the Project Establish the Proposed Research?

Yes, within the approved cross-sectional and predictive scope. RQ1 established a small adjusted association; RQ2 tested and did not establish meaningful incremental discrimination; RQ3 completed the prespecified model comparison and selected baseline XGBoost; and RQ4 demonstrated global and local explainability, including the supplementary explanatory role of PHQ-9. A null or negligible incremental effect is still a substantive research result because it prevents overclaiming the value of adding a psychological variable solely because its regression p-value is statistically significant.

Practical Significance

For population-health analytics, the findings argue for holistic but evidence-disciplined feature selection. Psychological measures may add interpretive context even when they do not improve discrimination. For model governance, the results also show why a single performance number is insufficient: the selected XGBoost model discriminated best but was poorly calibrated, and its sensitivity/specificity changed materially with threshold choice. Any future operational use would therefore require external validation, recalibration, fairness assessment, and a use-case-specific operating threshold.

Implementation and User Benefit

Deployment Approach

No clinical deployment was performed in this capstone. A future validated implementation could expose the preprocessing pipeline and calibrated model through a secured API, with a dashboard presenting the classification, calibrated probability,

confidence/uncertainty information, and SHAP explanation. The current XGBoost probabilities should not be used for this purpose without external validation and calibration. The most appropriate near-term implementation is therefore a research dashboard for cohort exploration and model explanation rather than an automated clinical decision system.

System Integration and User Interaction

In a validated environment, structured demographic and lifestyle fields could be mapped from a population-health database or electronic health record to the same feature definitions used during training. Users would see a model output together with the factors that pushed the prediction toward or away from healthy weight. PHQ-9 could be displayed as contextual psychological information, but the system should not imply that PHQ-9 alone determines weight status or that SHAP values are causal effects.

Benefits to Users

Researchers benefit from a reproducible baseline-versus-extended evaluation framework. Public-health teams can use the approach to compare multidimensional drivers and identify where additional data actually improve prediction. Healthcare organizations gain a model-governance example showing the need to examine discrimination, class-specific performance, calibration, and explainability together. The principal strategic benefit is reduced risk of deploying an apparently sophisticated model whose probabilities or added variables have not been shown to provide meaningful value.

Example Use Case

A public-health analytics team could apply the framework to a new, externally validated adult cohort to estimate healthy-weight classification using routinely collected demographic and lifestyle variables. A dashboard could show a calibrated model score and SHAP explanation,

while separately displaying PHQ-9 severity as contextual information. If PHQ-9 again fails to improve validated discrimination, the organization could avoid collecting or processing it solely for prediction while still recognizing its relevance to holistic care and mental-health assessment.

Limitations and Further Improvements

Limitations

Table 9

Study Limitations and Impact on Interpretation

Limitation	Impact on Interpretation
Cross-sectional design	Temporal or causal direction between depressive symptoms and weight cannot be established.
Self-reported predictors	Activity, sleep, dietary recall, and PHQ-9 can contain recall/reporting error.
Survey design not incorporated in modelling	The predictive analysis does not provide nationally representative weighted effect estimates.
Class imbalance	Only 24.35% were healthy weight; threshold-dependent metrics can favor the majority class.
Single NHANES cycle	Findings may not generalize to other periods, countries, or clinical populations.
Calibration	Selected XGBoost discriminated best but had poor probability calibration.
Model-family scope	The study intentionally evaluated LR, RF, and XGBoost; other architectures were outside the approved capstone scope.
PHQ-9 interpretation	PHQ-9 measures depressive symptom severity and is not a standalone clinical diagnosis in this analysis.
SHAP interpretation	SHAP explains model behaviour, not causal effects or biological mechanisms.

Note. These limitations define the boundaries of interpretation and motivate the future

improvements described in the following section.

Future Improvements

Immediate methodological improvements include external validation on another NHANES cycle, survey-weighted sensitivity analyses for population inference, formal recalibration using a separate validation sample, subgroup/fairness analysis, temporal or longitudinal data, richer psychosocial variables, and prospectively defined operating thresholds.

Repeated or nested resampling could further quantify model-selection uncertainty. A future clinical implementation would also require governance, privacy, monitoring, and human oversight.

Future Scope and Third-Year Dissertation

The present capstone deliberately stops at the questions supported by the available cross-sectional data and the approved ten-week scope. The third-year dissertation can extend this work into a more advanced research programme rather than simply adding more algorithms to the same dataset. Potential directions include multi-cycle or longitudinal analysis of the temporal depression-weight relationship; richer psychological constructs such as anxiety, stress, social determinants, and medication context; external validation across populations; causal-inference designs where assumptions can be defended; and calibrated, fairness-audited prediction.

A further dissertation extension could investigate a governed large language model or agentic layer that translates validated statistical and SHAP outputs into personalized, evidence-linked explanations or recommendations. Such a system would need strict separation between model evidence and generated language, retrieval from approved clinical/public-health guidance, safety constraints, and human review. No such LLM/agentic component is claimed as part of the current results. Everything required to answer the four present research questions is included in this final report; the dissertation would represent a new research phase.

Conclusion

This study developed and completed a reproducible framework for assessing whether depressive symptom severity adds predictive value to healthy-weight classification beyond conventional demographic and lifestyle information. Using 4,657 adults from NHANES 2017-2018, the analysis combined inferential statistics, Logistic Regression, Random Forest,

XGBoost, paired-bootstrap model comparison, threshold robustness, calibration analysis, and SHAP explainability.

PHQ-9 was a relevant variable to test because it is a validated measure of depressive symptom severity and prior longitudinal research supports a depression-obesity relationship. The analysis confirmed a statistically significant but small adjusted association: each one-point increase in PHQ-9 corresponded to approximately 2.7% lower adjusted odds of healthy-weight status. However, adding PHQ-9 did not materially improve held-out discrimination in any model family. Baseline XGBoost achieved the highest held-out ROC-AUC of 0.7109 and was selected, while subsequent calibration analysis showed that its raw probabilities should not be interpreted as reliable individual probabilities without recalibration.

SHAP added a complementary insight: PHQ-9 ranked seventh in the extended XGBoost model and could meaningfully shift some individual predictions even though it added almost no population-level discrimination. The project therefore demonstrates that statistical association, explanatory contribution, and incremental predictive utility are different evidentiary questions. Within the current scope, all four research questions were answered. The results support psychological health as part of a multifactorial understanding of weight, but they do not support PHQ-9 as a strong standalone predictive enhancement. Future dissertation research can extend the framework with longitudinal data, external validation, richer psychosocial information, calibration and fairness analysis, and potentially a governed LLM-assisted explanation layer.

References

- Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82-115.
<https://doi.org/10.1016/j.inffus.2019.12.012>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32.
<https://doi.org/10.1023/A:1010933404324>
- Centers for Disease Control and Prevention. (2024). Adult obesity facts.
<https://www.cdc.gov/obesity/>
- Chaput, J.-P., Dutil, C., Featherstone, R., Ross, R., Giangregorio, L., Saunders, T. J., Janssen, I., Poitras, V. J., Kho, M. E., Ross-White, A., & Carrier, J. (2020). Sleep duration and health in adults: An overview of systematic reviews. *Applied Physiology, Nutrition, and Metabolism*, 45(10 Suppl. 2), S218-S231. <https://doi.org/10.1139/apnm-2020-0034>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785-794). Association for Computing Machinery.
<https://doi.org/10.1145/2939672.2939785>
- Esteva, A., Robicquet, A., Ramsundar, B., Kuleshov, V., DePristo, M., Chou, K., Cui, C., Corrado, G., Thrun, S., & Dean, J. (2019). A guide to deep learning in healthcare. *Nature Medicine*, 25(1), 24-29. <https://doi.org/10.1038/s41591-018-0316-z>
- Hosmer, D. W., Jr., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.

- Hruby, A., & Hu, F. B. (2015). The epidemiology of obesity: A big picture. *Pharmacoeconomics*, 33(7), 673-689. <https://doi.org/10.1007/s40273-014-0243-x>
- Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence* (Vol. 2, pp. 1137-1143). Morgan Kaufmann.
- Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2001). The PHQ-9: Validity of a brief depression severity measure. *Journal of General Internal Medicine*, 16(9), 606-613. <https://doi.org/10.1046/j.1525-1497.2001.016009606.x>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765-4774.
- Luppino, F. S., de Wit, L. M., Bouvy, P. F., Stijnen, T., Cuijpers, P., Penninx, B. W. J. H., & Zitman, F. G. (2010). Overweight, obesity, and depression: A systematic review and meta-analysis of longitudinal studies. *Archives of General Psychiatry*, 67(3), 220-229. <https://doi.org/10.1001/archgenpsychiatry.2010.2>
- Mannan, M., Mamun, A., Doi, S., & Clavarino, A. (2016). Prospective associations between depression and obesity for adolescent males and females: A systematic review and meta-analysis of longitudinal studies. *PLOS ONE*, 11(6), e0157240. <https://doi.org/10.1371/journal.pone.0157240>
- Mozaffarian, D., Hao, T., Rimm, E. B., Willett, W. C., & Hu, F. B. (2011). Changes in diet and lifestyle and long-term weight gain in women and men. *The New England Journal of Medicine*, 364(25), 2392-2404. <https://doi.org/10.1056/NEJMoa1014296>

- National Center for Health Statistics. (n.d.). NHANES 2017-2018: Data, documentation, codebooks, SAS code, and frequencies. Centers for Disease Control and Prevention. <https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?BeginYear=2017>
- Pereira-Miranda, E., Costa, P. R. F., Queiroz, V. A. O., Pereira-Santos, M., & Santana, M. L. P. (2017). Overweight and obesity associated with higher depression prevalence in adults: A systematic review and meta-analysis. *Journal of the American College of Nutrition*, 36(3), 223-233. <https://doi.org/10.1080/07315724.2016.1261053>
- Powers, D. M. W. (2011). Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation. *Journal of Machine Learning Technologies*, 2(1), 37-63.
- Rajkomar, A., Dean, J., & Kohane, I. (2019). Machine learning in medicine. *The New England Journal of Medicine*, 380(14), 1347-1358. <https://doi.org/10.1056/NEJMra1814259>
- Shickel, B., Tighe, P. J., Bihorac, A., & Rashidi, P. (2018). Deep EHR: A survey of recent advances in deep learning techniques for electronic health record analysis. *IEEE Journal of Biomedical and Health Informatics*, 22(5), 1589-1604. <https://doi.org/10.1109/JBHI.2017.2767063>
- Swift, D. L., Johannsen, N. M., Lavie, C. J., Earnest, C. P., & Church, T. S. (2014). The role of exercise and physical activity in weight loss and maintenance. *Progress in Cardiovascular Diseases*, 56(4), 441-447. <https://doi.org/10.1016/j.pcad.2013.09.012>
- Topol, E. J. (2019). High-performance medicine: The convergence of human and artificial intelligence. *Nature Medicine*, 25(1), 44-56. <https://doi.org/10.1038/s41591-018-0300-7>
- World Health Organization. (2024). Obesity and overweight. <https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight>

Yu, K.-H., Beam, A. L., & Kohane, I. S. (2018). Artificial intelligence in healthcare. *Nature Biomedical Engineering*, 2(10), 719-731. <https://doi.org/10.1038/s41551-018-0305-z>

Appendix A: Expanded Analytical Data Dictionary and Reproducibility Notes

The final modelling architecture used the 15 variables listed in the Materials and Methods section, with HEALTHY_WEIGHT as the target. Source BMI variables (BMXBMI and BMI_CATEGORY) were retained only for cohort construction/outcome derivation and excluded from predictors. Training-only median imputation was used for continuous predictors; training-only most-frequent imputation was used for categorical predictors; continuous variables were standardized and categorical variables one-hot encoded. The final reproducibility checks confirmed N = 4,657, training N = 3,725, test N = 932, 14 baseline predictors, 15 extended predictors, PHQ9_TOTAL as the sole extension, six model specifications, and baseline XGBoost as the selected model.

Table 10

Expanded Analytical Data Dictionary

Variable	Analytical Treatment
RIDAGEYR	Age at screening; Baseline
RIAGENDR	Sex; Baseline
RIDRETH3	Race/Hispanic origin; Baseline
DMDEDUC2_CLEAN	Educational attainment; Baseline
DMDMARTL_CLEAN	Marital status; Baseline
INDFMPIR	Family income-to-poverty ratio; Baseline
PAQ605_CLEAN	Vigorous work activity; Baseline
PAQ620_CLEAN	Moderate work activity; Baseline
PAQ650_CLEAN	Vigorous recreational activity; Baseline
PAQ665_CLEAN	Moderate recreational activity; Baseline
PAD680_CLEAN	Sedentary minutes/day; Baseline
SLD012	Typical sleep duration; Baseline
DR1TKCAL	Day-1 total energy intake; Baseline
DR1TFIBE	Day-1 dietary fibre; Baseline
PHQ9_TOTAL	Depressive symptom severity, 0-27; Extended only

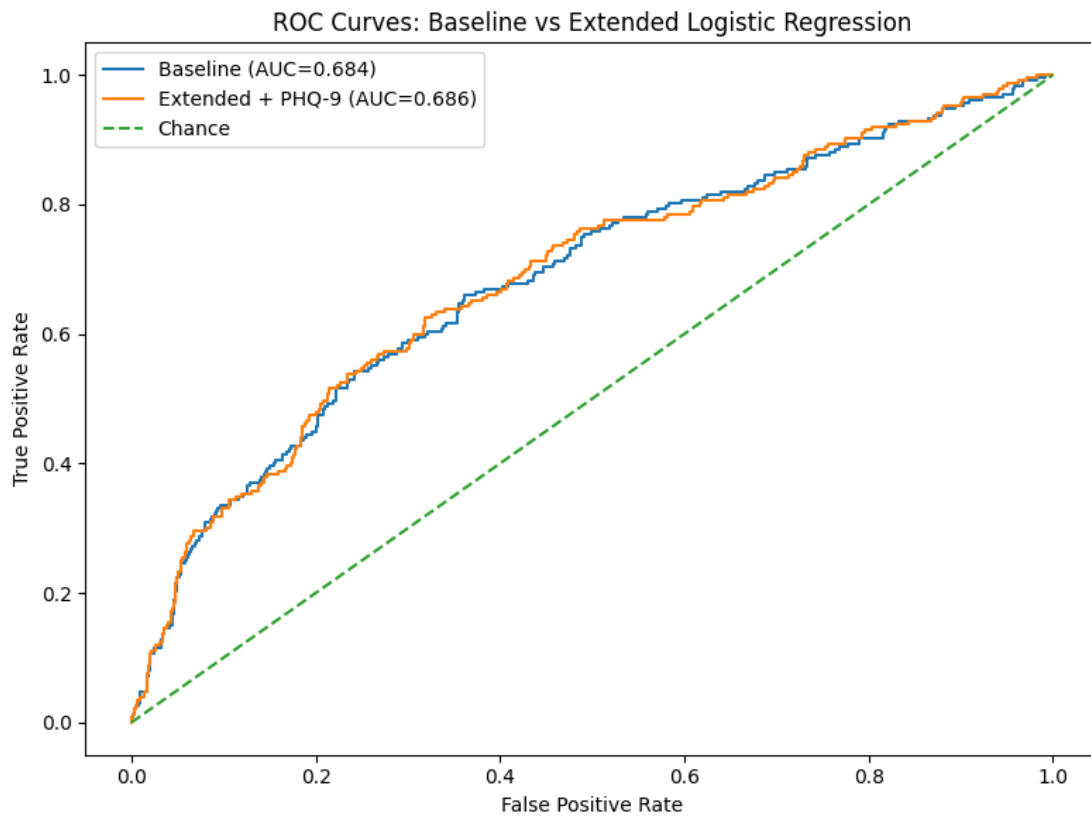
Note. BMI-derived variables were excluded from predictors to prevent target leakage; PHQ9_TOTAL was the only predictor added to the extended specification.

Appendix B: Additional Notebook Figures

The following figures are direct graphical outputs from the fully executed Python notebook and supplement the figures retained in the main report.

Figure B1

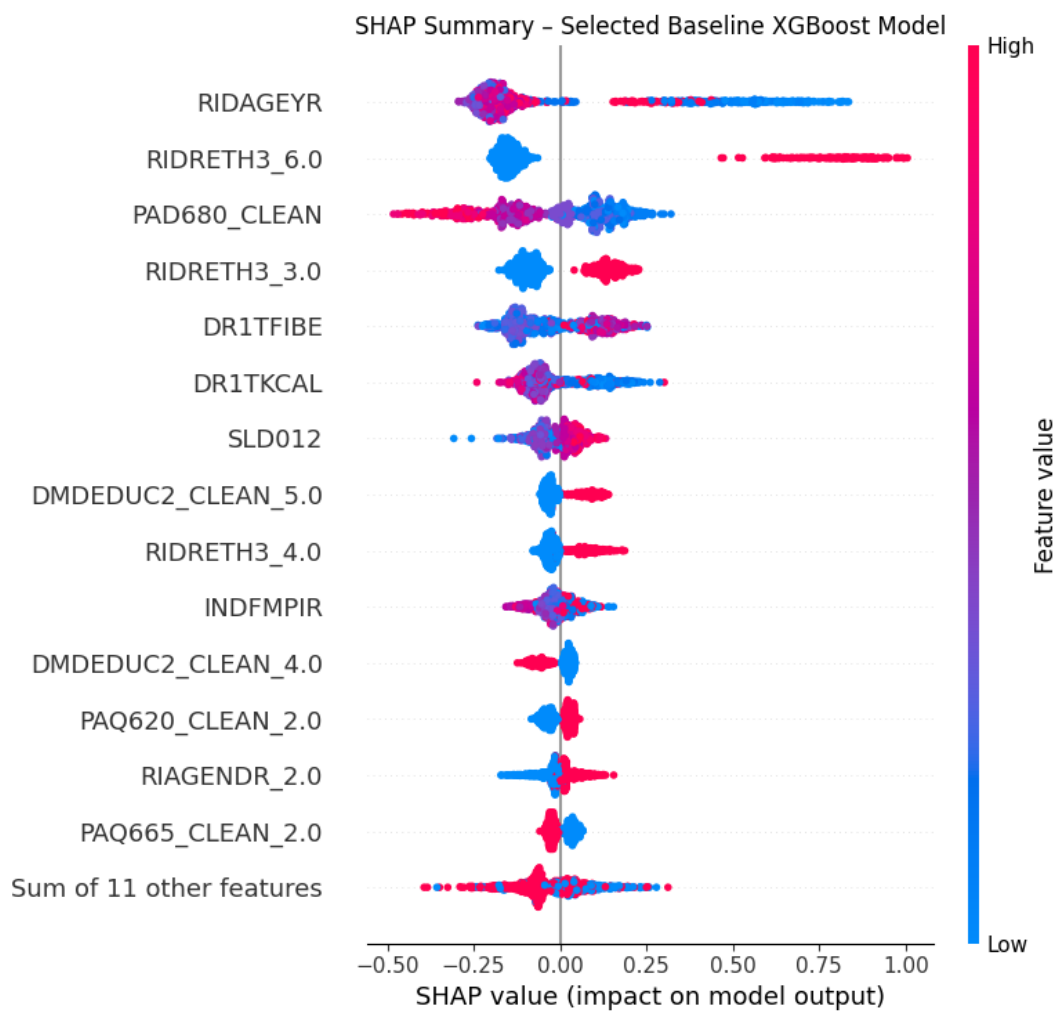
ROC Curves for Baseline vs Extended Logistic Regression



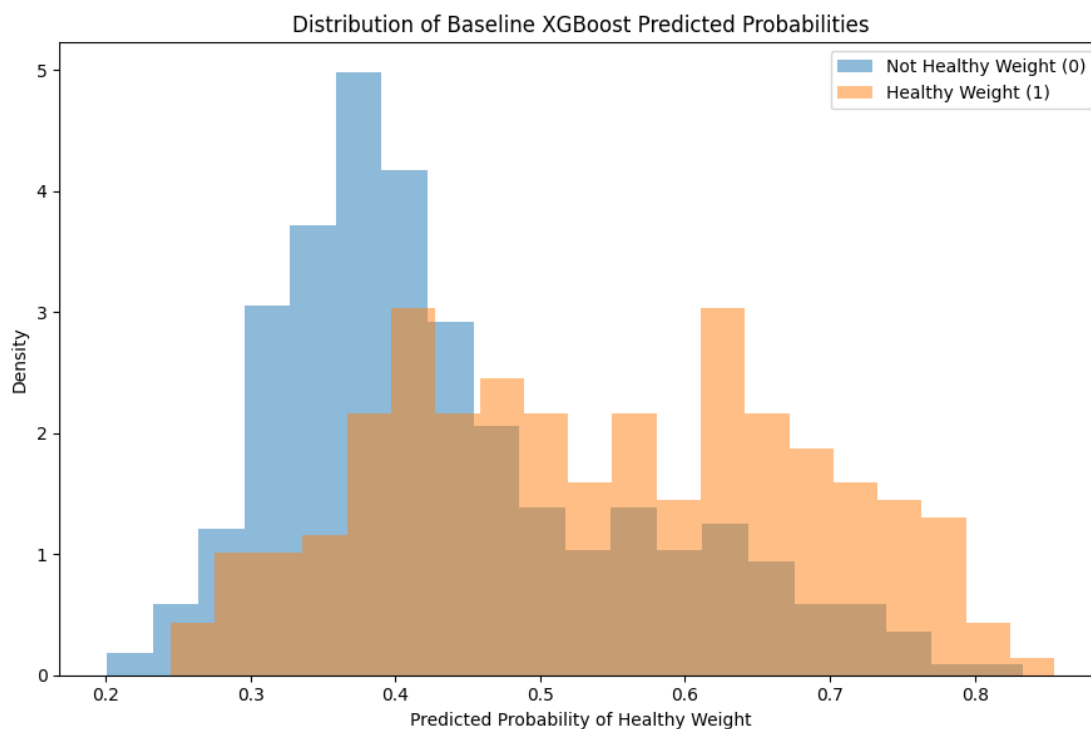
Note. The nearly overlapping ROC curves visually reinforce the negligible incremental discrimination from adding PHQ-9 to Logistic Regression.

Figure B2

SHAP Summary for Selected Baseline XGBoost



Note. This is the primary SHAP model because baseline XGBoost had the highest held-out ROC-AUC.

Figure B3*Predicted-Probability Distributions*

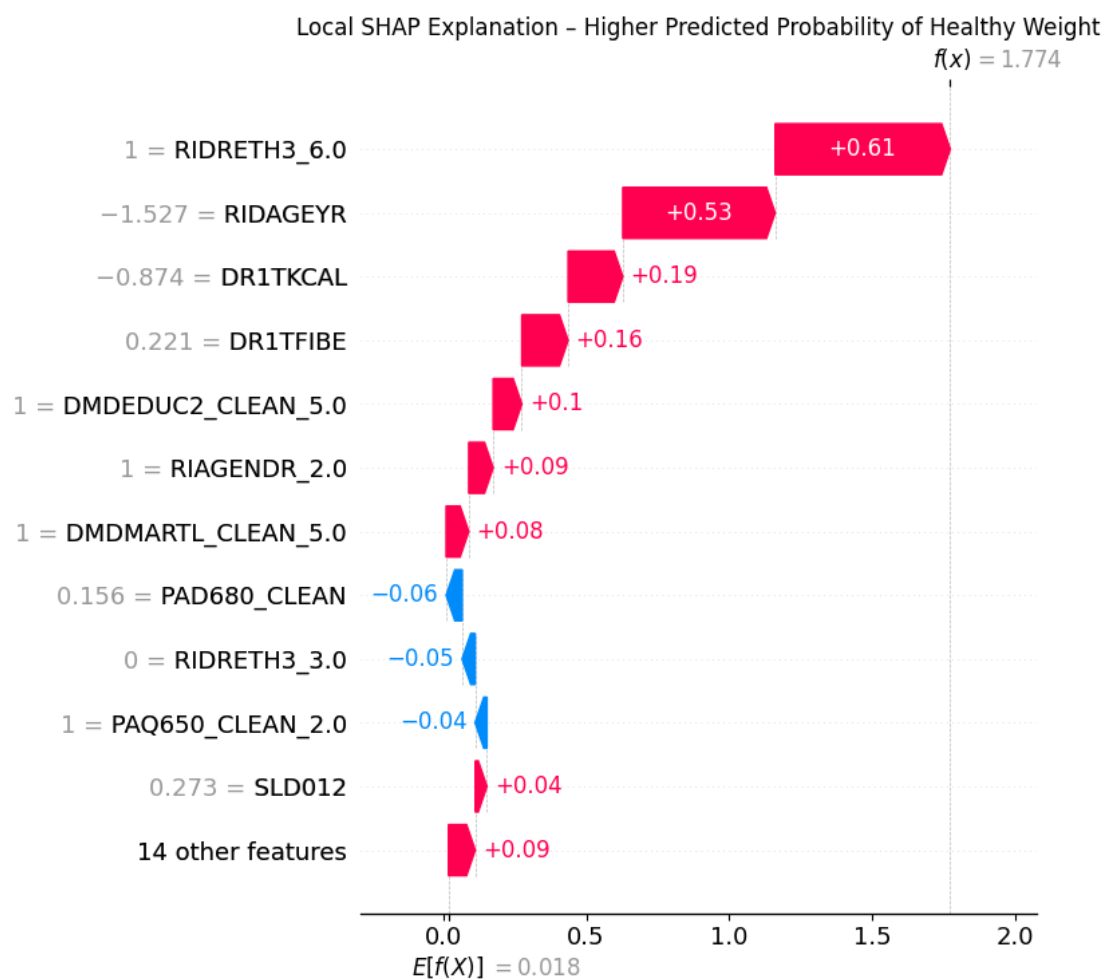
Note. The notebook distribution plot supplements the calibration analysis and helps explain the XGBoost model's tendency to assign higher probabilities than the observed healthy-weight prevalence.

Appendix C: Local SHAP Explanations

Two held-out participants were selected in the executed notebook to demonstrate individual-level model explanations. These examples are descriptive and should not be interpreted as causal explanations.

Figure C1

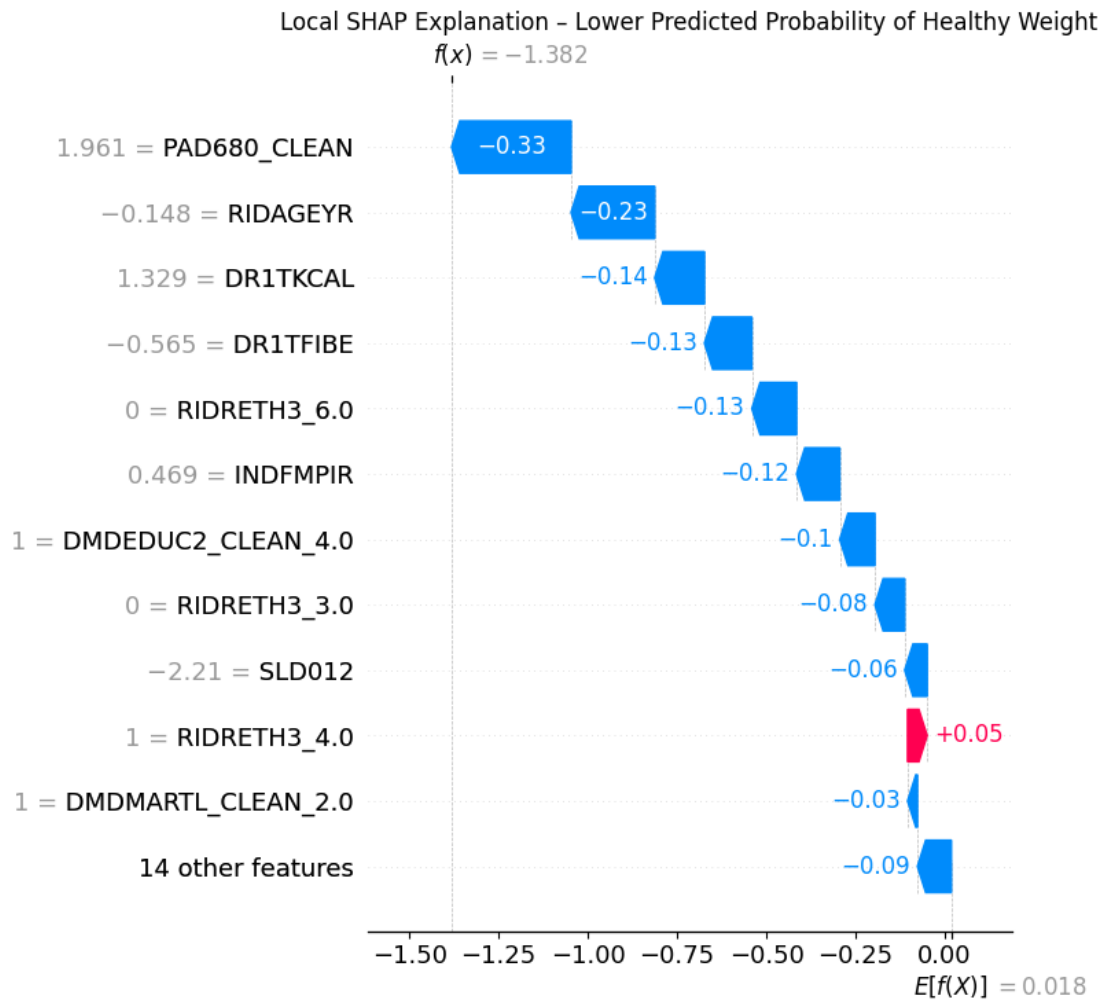
Local SHAP Explanation: High Healthy-Weight Probability



Note. The selected participant had predicted probability 0.8549 and actual healthy-weight class 1. The waterfall plot shows the features that moved the prediction from the model baseline toward the final output.

Figure C2

Local SHAP Explanation: Low Healthy-Weight Probability



Note. The selected participant had predicted probability 0.2007 and actual class 0. The waterfall plot illustrates how a different feature combination shifts the prediction away from healthy weight.

Appendix D: Notebook Audit Trail

The fully executed notebook contains the complete source code and visible outputs for cohort construction, missing-data audit, EDA, predictor refinement, inferential testing, train-test splitting, leakage-safe preprocessing, model tuning, held-out comparison, SHAP, incremental-value bootstrapping, threshold robustness, calibration, research-question synthesis, and final reproducibility checks. Selected screenshots are embedded throughout the main report and Appendices B-C; the complete executable record is maintained in the project repository.